

HW 0

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Chapter 1

1. One potential edge case that wasn't defined in the book is nested classes. For example, if you were to make a Linked List class, the Node class may be an inner class that doesn't need to exist outside of the Linked List class scope.

Attempting to use inner classes actually leads to a compile error:

```
class LinkedListStack {
    class Node {
        init(data, next) {
            self.data = data;
            self.next = next;
        }
    }

    init() {
        self.head = nil;
    }
    // ...
}
```

[line 2] Error at 'class ': Expect method name.

The correct way to implement this is to write:

```
class Node {
    init(data, next) {
        this.data = data;
        this.next = next;
    }
}

class LinkedListStack {

    init() {
        this.head = nil;
    }

    push(data) {
        if (this.head == nil) {
            this.head = Node(data, nil);
        }
        else {
```

```

        var next = Node(data, this.head);
        this.head = next;
    }
}

pop() {
    if (this.head == nil) {
        return nil;
    }
    else {
        var out = this.head;
        this.head = this.head.next;
        return out;
    }
}
}

```